DO NOW, AO1 REVISION

Solar System, stars, orbits and satellites

Space physics
Paper 2, Separate Physics only
25 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Name the main objects in our Solar System.**	The Sun, eight planets, dwarf planets, moons, asteroids and comets	☐
2	**What is a galaxy?**	A huge collection of stars held together by gravity.	☐
3	**Give the name of our galaxy.**	The Milky Way	☐
4	**Name a cloud of dust and gas where stars form.**	A nebula	☐
5 ★	**Name the force that pulls a nebula together to form a star.**	Gravity	☐
6	**What is a protostar?**	A hot ball of gas formed as gravity pulls a nebula together, before fusion starts.	☐
7 ★	**Name the process that begins when a protostar becomes hot enough.**	Nuclear fusion	☐
8 ★	**Explain why a main sequence star is stable.**	The inward pull of gravity is balanced by the outward push from fusion energy.	☐
9	**State what a Sun-sized star becomes after its main sequence stage.**	A red giant, then a white dwarf, then a black dwarf	☐
10	**State what a star much bigger than the Sun becomes after its main sequence stage.**	A red super giant, then a supernova, then a neutron star or a black hole	☐
11 ★	**State how elements up to iron are formed.**	By nuclear fusion in stars.	☐
12 ★	**State how elements heavier than iron are formed.**	In supernovae	☐

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	QUESTION	ANSWER	✓
13 ★	**State what keeps one object in orbit around another.**	Gravity	☐
14	**What is an orbit?**	The curved path of one object around another in space.	☐
15	**Name a natural satellite of Earth.**	The Moon	☐
16	**What is an artificial satellite?**	A human-made object in orbit around a planet.	☐
17 ★	**Explain why an object in orbit is accelerating.**	Its direction, and so its velocity, is always changing.	☐
18 ★	**Explain why the velocity of an object in a circular orbit changes but its speed does not.**	Gravity acts at right angles to its motion, changing its direction but not its speed.	☐
19 ★	**State how the speed of a stable orbit depends on its radius.**	The smaller the radius, the greater the speed.	☐
20	**State what the Sun was formed from.**	A nebula (a cloud of dust and gas) pulled together by gravity.	☐
21	**Name the natural satellites of planets.**	Moons	☐
22 ★	**List the stages in the life cycle of a Sun-sized star.**	Nebula, protostar, main sequence star, red giant, white dwarf, black dwarf	☐
23 ★	**List the stages in the life cycle of a star much bigger than the Sun.**	Nebula, protostar, main sequence star, red super giant, supernova, then a neutron star or a black hole	☐
24 ★	**State how elements made in stars are spread through the Universe.**	By supernova explosions.	☐
25	**What do planets orbit? What do moons and artificial satellites orbit?**	Planets orbit the Sun. Moons and artificial satellites orbit planets.	☐

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